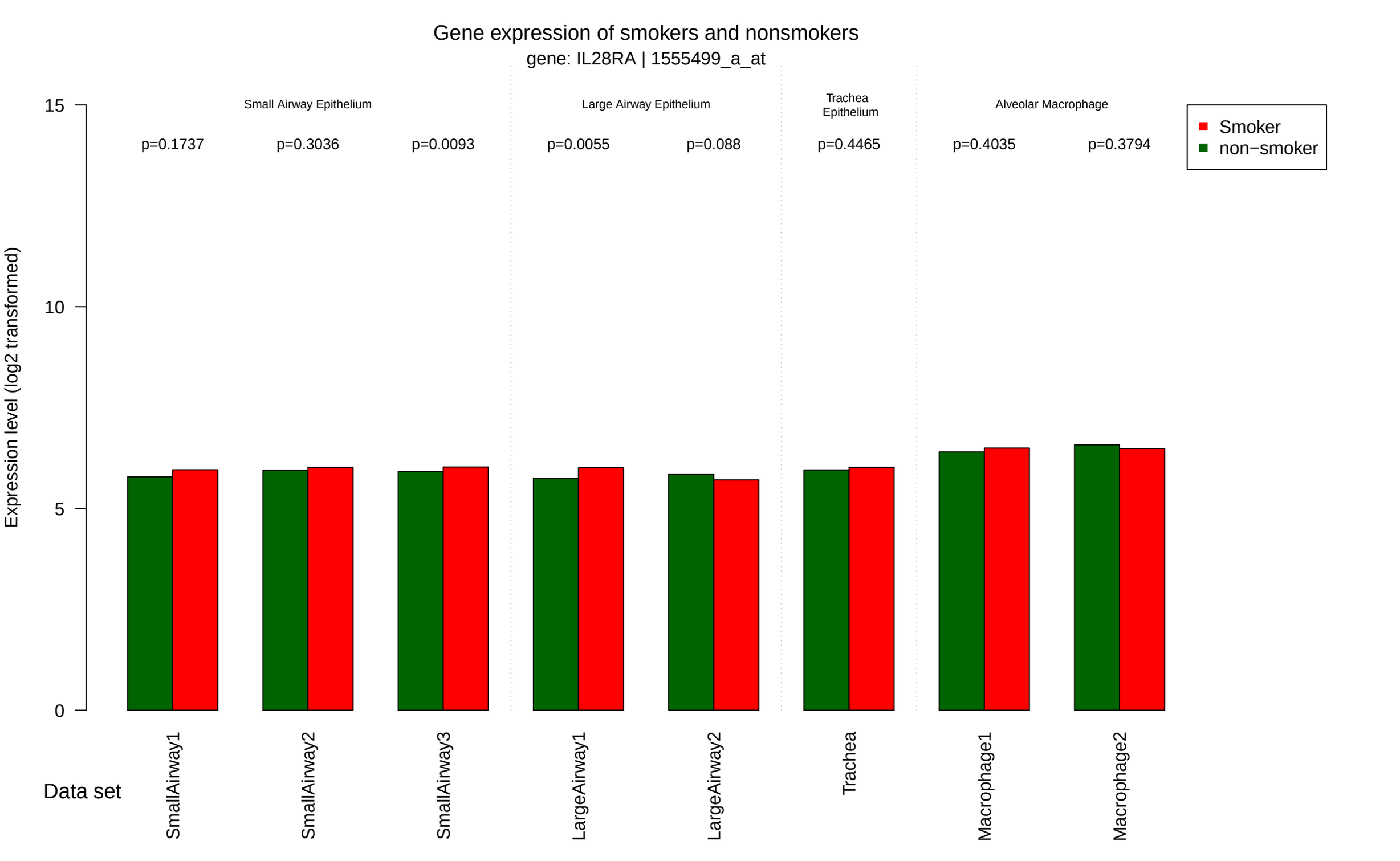
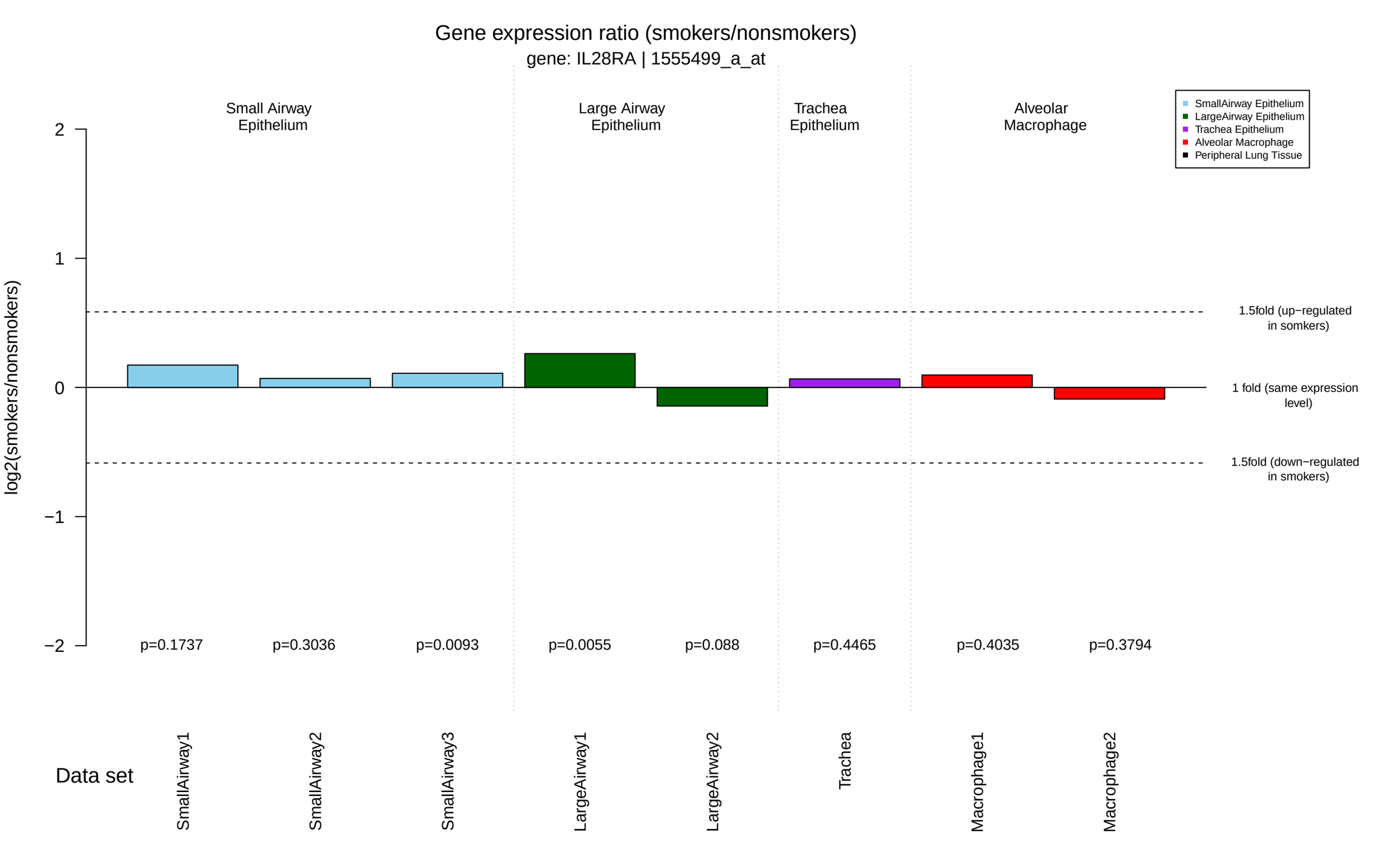
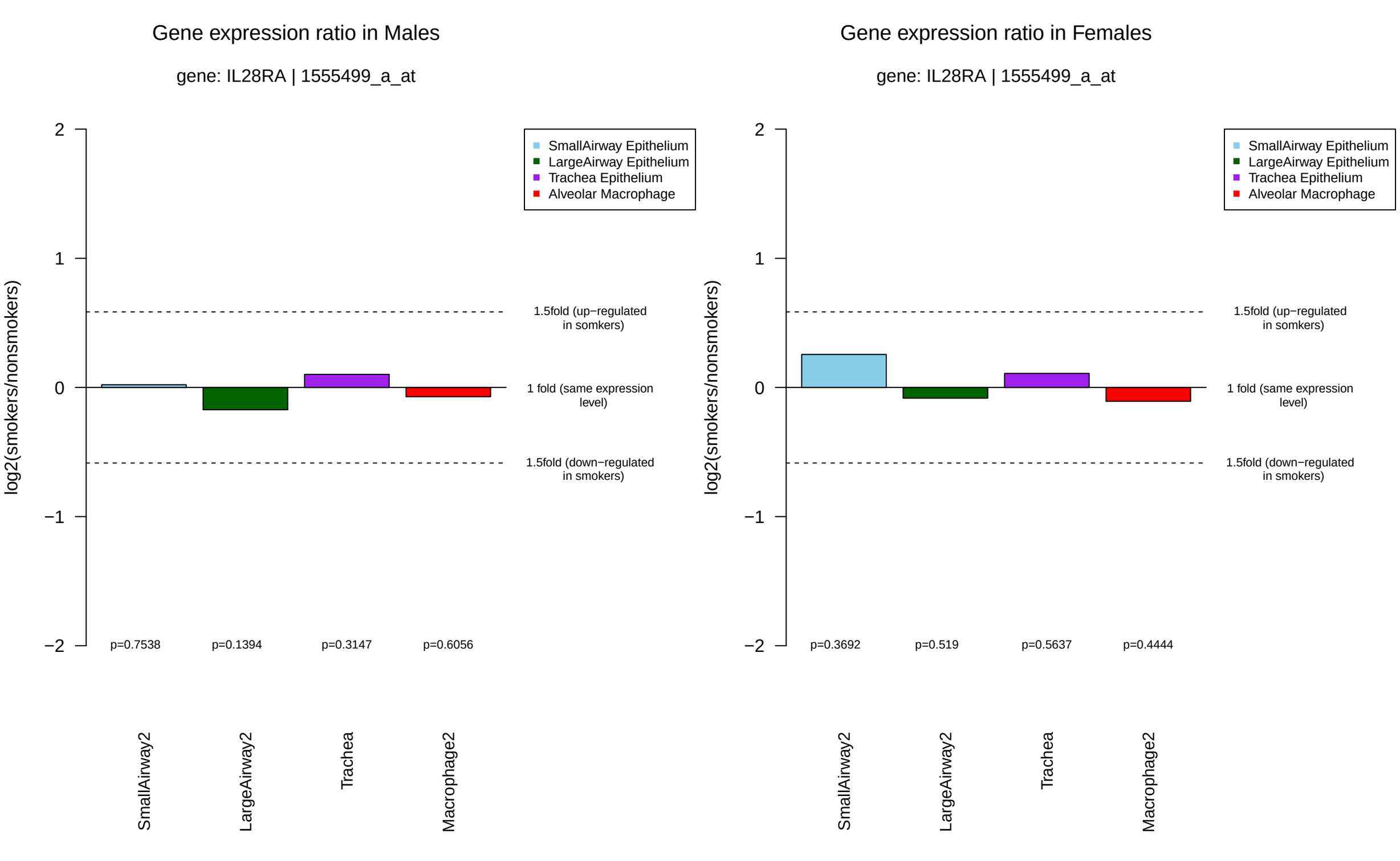
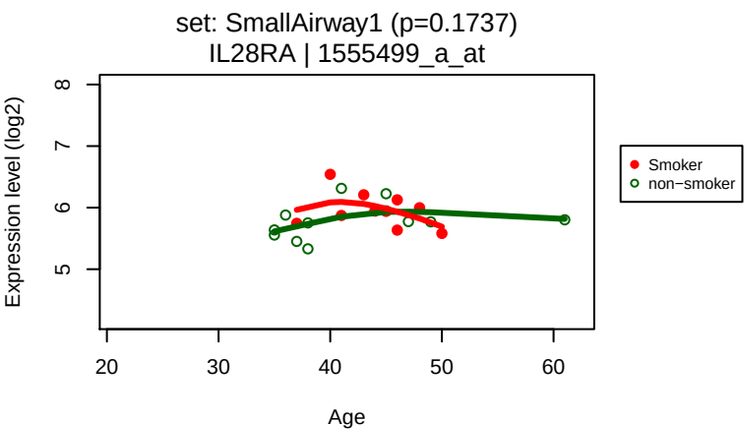
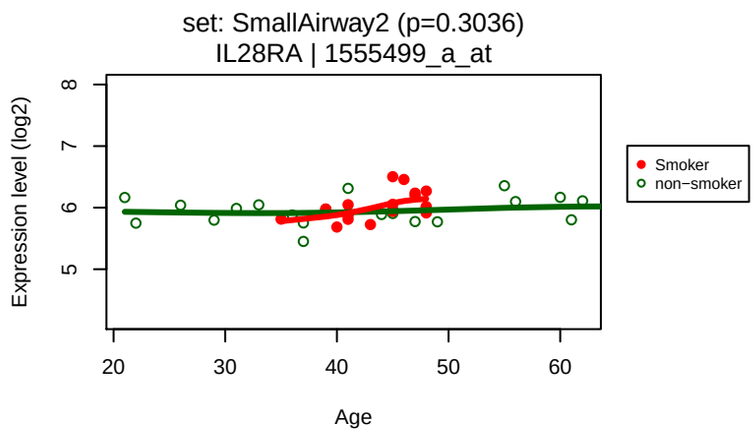
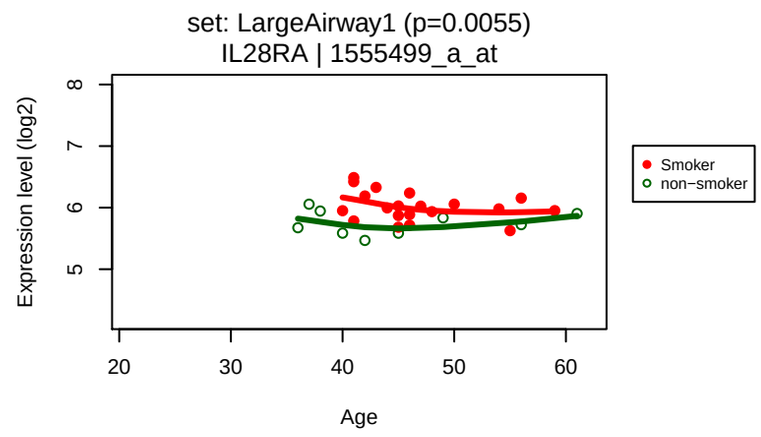

Gene expression of smokers and nonsmokers
gene: IL28RA | 1555499_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1737
p=0.3036
p=0.0093
p=0.0055
p=0.088
p=0.4465
p=0.4035
p=0.3794
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

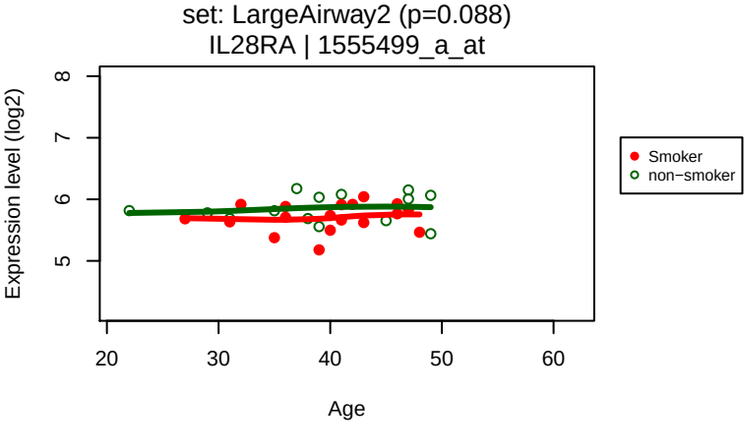

Gene expression ratio (smokers/nonsmokers)
gene: IL28RA | 1555499_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1737
p=0.3036
p=0.0093
p=0.0055
p=0.088
p=0.4465
p=0.4035
p=0.3794
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

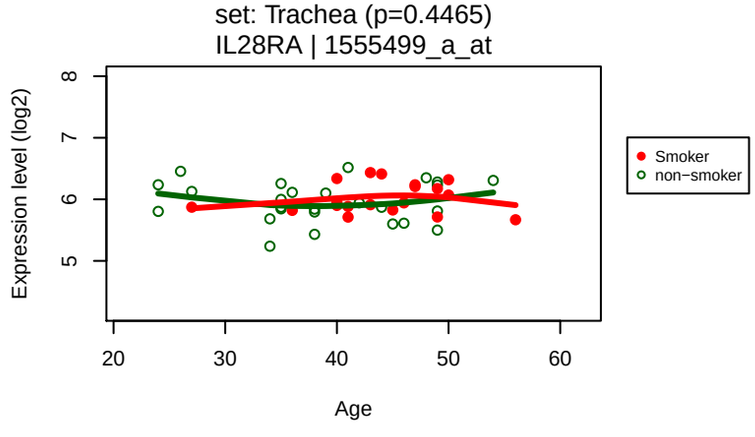

Gene expression ratio in Males
gene: IL28RA | 1555499_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7538
p=0.1394
p=0.3147
p=0.6056
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IL28RA | 1555499_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3692
p=0.519
p=0.5637
p=0.4444
SmallAirway2
LargeAirway2
Trachea
Macrophage2

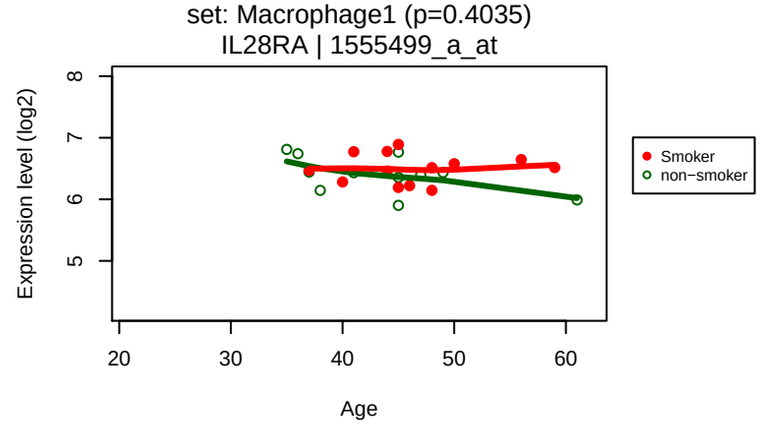

set: SmallAirway1 (p=0.1737)
IL28RA | 1555499_a_at
Expression level (log2)
Age
Smoker
non−smoker

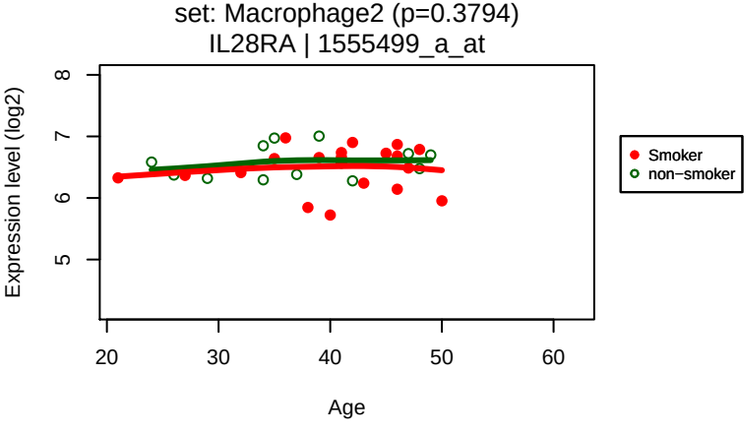

set: SmallAirway2 (p=0.3036)
IL28RA | 1555499_a_at
Expression level (log2)
Age
Smoker
non−smoker

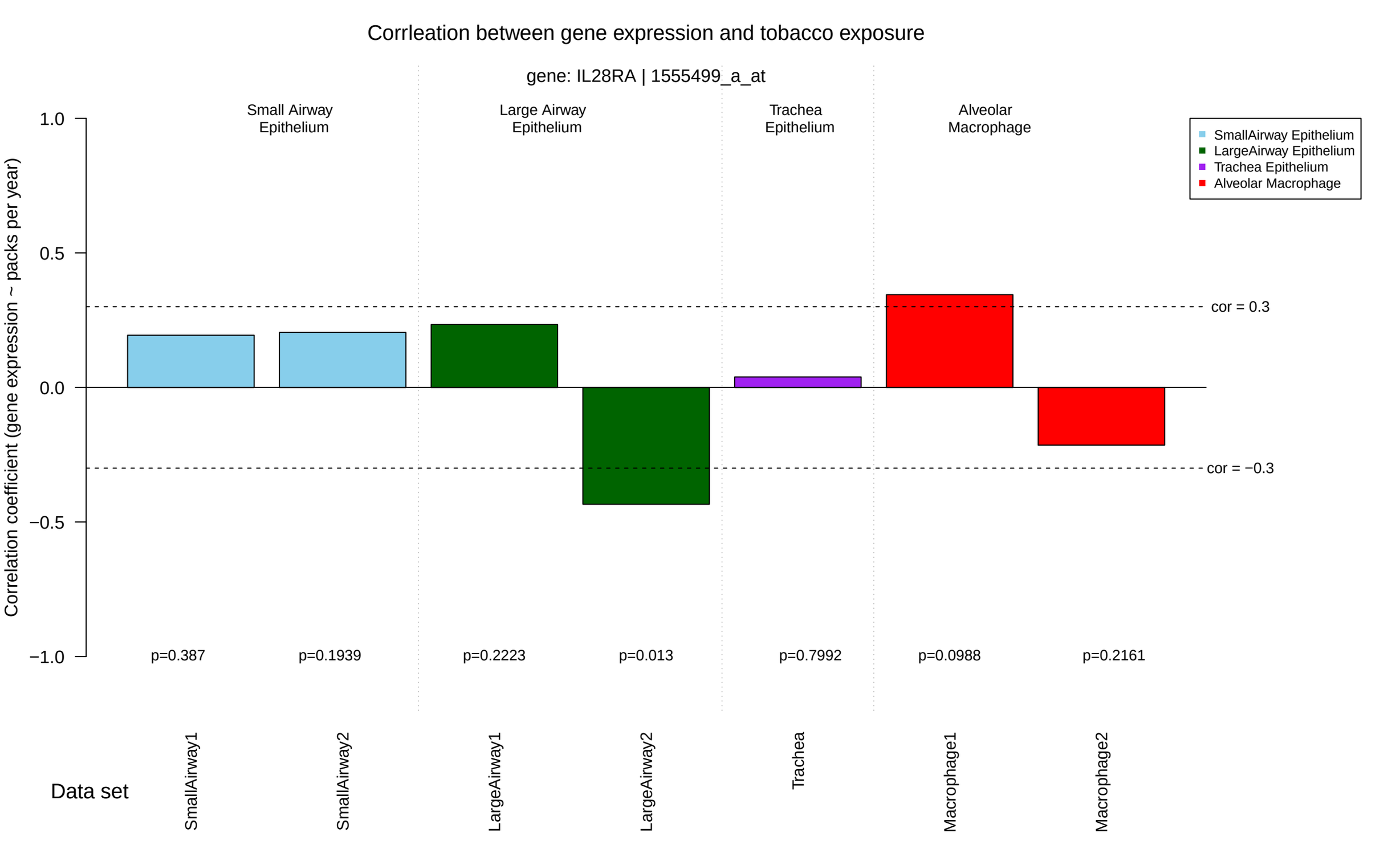

set: LargeAirway1 (p=0.0055)
IL28RA | 1555499_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.088)
IL28RA | 1555499_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4465)
IL28RA | 1555499_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4035)
IL28RA | 1555499_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3794)
IL28RA | 1555499_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IL28RA | 1555499_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.387
p=0.1939
p=0.2223
p=0.013
p=0.7992
p=0.0988
p=0.2161
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2